# CCTV-BASED AI INTRUSION DETECTION SYSTEM

Khor Soon Keat
*Faculty of Artificial Intelligence and Cyber Security (FAIX)*
*Universiti Teknikal Malaysia Melaka (UTeM)*
*Melaka, Malaysia*
Ts. Dr. Abdul Syukor bin Mohamad Jaya
*Faculty of Artificial Intelligence and Cyber Security (FAIX)*
*Universiti Teknikal Malaysia Melaka (UTeM)*
*Melaka, Malaysia*

**Abstract—**This paper presents SecureVision, a CCTV-based AI intrusion detection prototype for indoor laboratory security. The system combines YOLOv8 person detection and tracking, a configurable restricted-zone polygon, MediaPipe face localization, FaceNet authorized-person recognition, multi-frame hybrid authorization, finite Entry/Exit monitoring, structured evidence records, and Telegram/email notification. A person-first decision pipeline ensures that only tracked human detections continue to authorization and intrusion processing. The tracked bottom-center point is used for both zone membership and line-crossing geometry. While identity evidence is incomplete, the system remains in a Checking Authorization state and blocks siren, database, and alert actions. A stable authorized identity is permitted without an intrusion alert; an unknown person who remains inside the protected polygon until the confirmation delay expires produces snapshot and video evidence, a SQLite event record, siren activation, and external notifications. The Flask application provides Admin and read-only Viewer dashboards, authorized-person management, attendance/access history, record archive/restore and export, settings, CCTV readiness checking, and controlled Face Angle, Distance, and Movement Speed evaluation. Stored labelled data produced 100.00% accuracy for Face Angle, 66.67% accuracy and 62.22% F1-score for Distance, and 95.76% accuracy with 95.93% F1-score for Movement Speed. The prototype demonstrates an integrated and auditable AI-assisted CCTV workflow.

***Keywords—***CCTV, intrusion detection, YOLOv8, MediaPipe, FaceNet, restricted zone, attendance monitoring, Flask, SQLite, alert notification

## I. INTRODUCTION

### *A. Background and Motivation*

Laboratories contain computers, sensors, experimental devices, and learning materials that require reliable security monitoring. Traditional CCTV provides useful recorded footage, but it still depends on continuous human observation. When staff are not watching the relevant feed, unauthorized access may only be noticed after the footage is reviewed. Motion-only detection can also generate unnecessary alerts because it reacts to general image changes caused by animals, insects, shadows, lighting variation, or moving background objects.

SecureVision was developed as a person-centred CCTV security prototype. It supports webcam and Hikvision CCTV/RTSP input, detects and tracks people with YOLOv8, checks a configurable restricted-zone polygon, verifies registered people through MediaPipe and FaceNet, records authorized entry and exit events, stores intrusion evidence, and sends Telegram and email alerts. The security decision is therefore based on the detected object class, track continuity, physical position, face identity evidence, and confirmation time rather than a single motion event.

### *B. Problem Statement and Objectives*

The work addresses three related problems: delayed detection caused by manual CCTV observation, false alarms produced by non-human movement, and the absence of automatically organized intrusion and access records. The corresponding objectives are to: (1) detect human presence in restricted laboratory areas using YOLOv8; (2) identify authorized persons using MediaPipe face detection and FaceNet recognition; and (3) implement an access and intrusion management workflow that records attendance/access events, captures evidence, manages records, and sends alert notifications.

### *C. Main Contributions*

The contribution is an integrated prototype rather than an isolated detector. First, a person-only gate prevents selected non-human detections from entering the intrusion decision path. Second, a Checking Authorization state and multi-frame rule prevent a weak one-frame identity decision from immediately allowing or alarming a person. Third, the same tracked bottom-center geometry supports restricted-zone membership and finite Entry/Exit crossing. Finally, every confirmed event is linked to evidence, database metadata, review status, archive/restore, export, and notification functions, making the workflow auditable.

## II. RELATED WORK

### *A. Intelligent Surveillance and Human Detection*

Object detection has progressed from handcrafted human descriptors such as HOG [6] to deep detectors and single-stage YOLO models [1]–[5], [7]. YOLO is suitable for real-time surveillance because a single network predicts object locations and classes with low latency. Intelligent surveillance surveys emphasize that detection, tracking, temporal reasoning, and usable event management must operate together in practical systems [8]. In SecureVision, only the person class proceeds to tracking, authorization, zone checking, and intrusion decisions; selected COCO non-human classes may support display or controlled filtering tests but do not create security events.

### *B. Face Recognition and Attendance*

FaceNet maps face images into a metric embedding space in which identity similarity can be measured [9]. MediaPipe provides efficient perception pipelines [10], while BlazeFace, Viola–Jones, MTCNN, and ArcFace illustrate alternative approaches to face localization and identity representation [11]–[14]. Face recognition has also been applied to attendance systems [15]. A practical limitation is that recognition quality depends on pose, distance, blur, illumination, occlusion, and visible face size. For this reason,

SecureVision combines face localization, FaceNet distance evidence, and multi-frame identity voting instead of confirming authorization from one frame.

### C. Intrusion Detection, Anomaly Analysis, and False Alarms

Surveillance research includes anomaly detection in real-world video [16], temporal regularity modelling [17], intrusion detection in video surveillance [18], and false-alarm reduction using learned feature distributions [19]. Background-modelling surveys also show why general foreground changes can be unreliable as direct security decisions [20]. Recent reviews describe the opportunities and deployment challenges of deep-learning video anomaly detection [21]. SecureVision takes a narrower but explainable approach: it first requires a tracked person, then applies identity, geometry, and dwell-time rules before an intrusion is confirmed.

### D. System Integration Technologies

OpenCV supports frame acquisition, overlays, snapshot capture, and video evidence [22]. Flask provides the browser-based application layer [23], SQLite provides lightweight local structured storage [24], and Telegram Bot API together with SMTP supports remote notification [25], [26]. These components connect a live AI decision to a reviewable record and evidence path.

TABLE I. COMPARISON OF MONITORING APPROACHES

| Approach | Human Gate | Identity | Records / Alerts |
|---|---|---|---|
| Traditional CCTV | Manual | No | Video only |
| Motion detector | No | No | Basic alert |
| Face attendance | Limited | Yes | Attendance |
| SecureVision | YOLOv8 | FaceNet hybrid | Evidence + alerts |

### E. Multi-Object Tracking and Identity Continuity

Tracking-by-detection methods show that reliable video analysis depends not only on detecting a person but also on associating that person across consecutive frames. SORT combines detections with motion prediction and data association for efficient online tracking [27], while Deep SORT adds an appearance metric to reduce identity switches during occlusion [28]. ByteTrack associates both high- and low-confidence detections to recover partially occluded targets [29], and Tracktor reuses detector regression to propagate tracks [30]. These approaches support the design choice of binding authorization evidence, dwell time, line crossing, and alert state to a persistent track rather than to an isolated frame.

Tracking continuity remains sensitive to missed detections, overlap, scale variation, illumination change, and camera placement. Surveillance and anomaly-detection reviews describe the need to combine detection, temporal reasoning, and operational event handling rather than relying on one frame alone [7], [8], [21]. Background-modelling research also shows why general foreground change can be unreliable as a direct alarm trigger [20]. SecureVision therefore uses confirmation windows, deadbands, and track-bound state, although crowded scenes can still affect Attendance and Entry/Exit consistency.

### F. Face Recognition Robustness

FaceNet, MediaPipe, BlazeFace, Viola-Jones, MTCNN, and ArcFace represent different stages of face localization, alignment, and identity representation [9]-[14]. Attendance research further demonstrates the practical use of face recognition for identity-linked records [15]. Across these approaches, pose, visible face size, blur, occlusion, illumination, and distance remain important operating factors. This supports SecureVision's repeated identity voting and Checking Authorization state and also explains why the Distance evaluation produced weaker results than the controlled Face Angle evaluation.

### G. Operational Security and Data Handling

A complete CCTV workflow must protect its stored information as well as its live decision. SecureVision separates Admin and Viewer permissions, masks CCTV credentials, requires confirmation for destructive record actions, and links each confirmed intrusion to evidence and workflow metadata. OpenCV, Flask, SQLite, Telegram, and SMTP provide the implementation path from frame acquisition to local storage and external notification [22]-[26]. A production deployment should additionally use stronger credential storage, encrypted off-device backups, formal retention periods, audit review, and documented consent for face and attendance data.

### H. Dataset Coverage and Future Extension

The YOLO family and general object-detection literature demonstrate how detector architecture, training data, and scene diversity influence real-time recognition performance [1]-[7]. SecureVision uses a person-first gate so selected non-human detections do not enter the security-decision path. Nevertheless, local validation should cover laboratory lighting, reflections, camera height, partial bodies, animals, insects, and moving background objects instead of assuming that general benchmark performance transfers directly to the deployment site.

Future work can compare SORT, Deep SORT, ByteTrack, and Tracktor on labelled Entry/Exit sequences [27]-[30], test more identities and repeated sessions, and measure identity-switch and crossing errors separately from person-detection accuracy. Additional evaluation should include longer distances, partial occlusion, network interruption, and multi-person overlap. Face anti-spoofing and multi-camera identity handoff can also be considered after the current single-source prototype is validated with a broader dataset.

### I. Research Gap and Proposed Position

Existing approaches frequently concentrate on one task: object detection, motion analysis, anomaly scoring, face recognition, or attendance. A detector can identify a person but does not by itself determine whether the person is authorized, whether the physical location is protected, or whether the observation should become a reviewable event. Similarly, a face-attendance application may record identity but may not provide restricted-zone logic, evidence capture, intrusion delay, external alerts, or lifecycle management for security records.

SecureVision addresses this integration gap at prototype scale. It does not claim a new deep neural architecture; its novelty is the explainable orchestration of established AI components with track-bound rules and a complete operational workflow. The person-only gate, bottom-center geometry, pending authorization state, confirmation delay, separate authorized/unknown outcomes, evidence linkage, and data-driven evaluation provide a coherent bridge from live detection to security action. This position also makes limitations visible because the effect of distance, tracking, face quality, and configuration can be evaluated separately.

## III. SYSTEM METHODOLOGY AND DECISION LOGIC

### *A. Development Methodology*

The project follows an Intelligent System Development Methodology with seven phases: problem identification and requirements analysis, literature review, system design, intelligent-model integration, implementation, testing and evaluation, and documentation. The method supports the combined development of the real-time computer-vision pipeline, camera configuration, web interfaces, database, explicit security rules, and evaluation evidence.

### *B. End-to-End Processing Flow*

OpenCV reads a webcam or CCTV/RTSP frame, after which YOLOv8 detects and tracks people. Each track enters two geometry branches. The Attendance/Access branch checks a configurable finite Entry/Exit line, whereas the Restricted Zone branch checks a protected polygon. MediaPipe and FaceNet provide face and identity evidence to a hybrid authorization stage. A crossing may be retained temporarily for the same track while identity is pending, but it is persisted only if the track becomes authorized within the verification window. An unknown track inside the zone starts a dwell timer; the intrusion workflow begins only if the track remains unknown and inside until the configured delay expires.

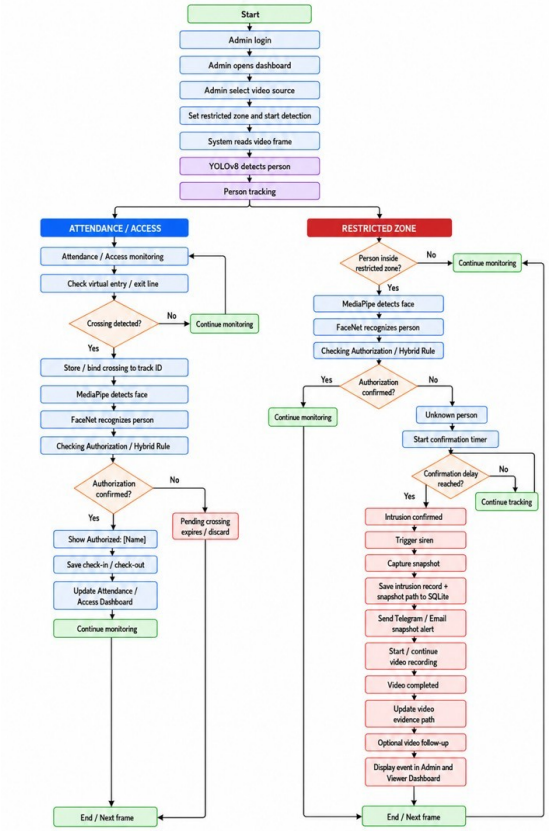



Fig. 1. Overall flow for Attendance/Access and Restricted Zone monitoring.

### *C. Geometry, Authorization, and Intrusion Rules*

For a tracked bounding box (x1, y1, x2, y2), the point used for floor-based decisions is the bottom-center or foot point:

$$p_f = ((x_1 + x_2)/2,\ y_2) \quad (1)$$

Polygon membership is evaluated for p_f rather than for the body centre. Entry/Exit direction is derived from the sign change of the point relative to the configured finite line, with a deadband and segment check to reject jitter and crossings outside the active line segment. The authorization state is based on repeated identity evidence. In general form, a track is authorized when its accepted FaceNet distance is below a configurable threshold and the same identity receives sufficient support over a recent frame window:

$$Authorized_i = [d_i < \tau_d] \wedge [votes_i \geq N_v] \quad (2)$$

The exact threshold and vote-window settings are configurable. Until (2) is satisfied, the display remains Checking Authorization and intrusion actions are blocked. For an unknown track, intrusion is confirmed only when person detection, zone membership, unknown identity, and dwell-time conditions are simultaneously true:

$$Intrusion_i = Person_i \wedge Inside_i \wedge Unknown_i \wedge (\Delta t_i \geq \tau_c) \quad (3)$$

This staged logic reduces one-frame recognition errors, brief zone touches, and non-human motion from becoming immediate security alarms.

TABLE II. MODEL AND DECISION PARAMETERS

| Component | Decision Rule | Purpose |
|---|---|---|
| YOLOv8 | Confidence gate | Accept person detections |
| FaceNet | Distance + votes | Stable identity evidence |
| Restricted zone | Foot point + delay | Reduce brief triggers |
| Entry/Exit | Line + deadband | Validate direction |

### *D. Per-Track State Management and Event Control*

The processing loop maintains security state per tracked person rather than using a single global face result. A newly detected person begins in a Detected state and advances to Checking Authorization when a usable face region is available. Stable identity evidence changes the state to Authorized; otherwise, an in-zone track remains under unknown-person observation until it leaves the polygon or reaches the confirmation delay. The track identifier binds the face votes, zone timer, pending line crossing, attendance identity, and alert suppression state to the same person.

Event control is important because a live camera produces many frames for the same physical event. Once an intrusion has been confirmed for a track, the evidence and notification workflow is not repeatedly recreated on every following frame. The event holds its own recording and database state until the configured workflow completes. In the attendance branch, a line crossing that occurs before authorization is retained only for the allowed verification window. It is committed after that same track becomes authorized; otherwise the pending crossing expires. These rules prevent duplicate alerts and reduce identity/crossing mismatch.

TABLE III. MAIN PER-TRACK STATES

| State | Entry Condition | Permitted Action |
|---|---|---|
| Detected | Person track created | Track and geometry |
| Checking | Face evidence pending | No siren or record |
| Authorized | Stable distance + votes | Allow + attendance |
| Unknown in zone | No valid identity | Run dwell timer |
| Confirmed | Delay reached | Evidence + alert |

### *E. Threat Model and System Scope*

The prototype considers an unauthorized human entering a camera-visible restricted laboratory region. The person may cross an access line, approach from outside the polygon, or remain briefly near its boundary. Authorized users are assumed to have registered face data and to be visible for enough frames to support recognition. Non-human motion, brief boundary contact, and incomplete one-frame identity evidence are treated as conditions that should not immediately create an intrusion. The system is designed to support an operator, not to replace physical access controls such as locks or door sensors.

The current scope is a local laboratory deployment with one active video source per monitoring process. It does not claim adversarial face-spoof resistance, guaranteed recognition under complete occlusion, or large-scale multi-camera identity handoff. Notification delivery depends on external network availability, while CCTV/RTSP acquisition depends on local network configuration. These boundaries are important when interpreting the controlled evaluation: the measurements validate the implemented pipeline under documented conditions, not every possible surveillance threat.

## IV. SYSTEM ARCHITECTURE AND IMPLEMENTATION

### *A. Architecture Overview*

The Flask application acts as the controller for authentication, dashboard routing, source selection, live streaming, system settings, authorized-person management, attendance/access records, intrusion records, alert functions, and testing pages. The processing layer combines OpenCV, YOLOv8, tracking, MediaPipe, FaceNet, geometry rules, and hybrid authorization. The output layer contains Admin and Viewer dashboards, siren and remote alerts, evidence files, attendance sessions, and SQLite records.

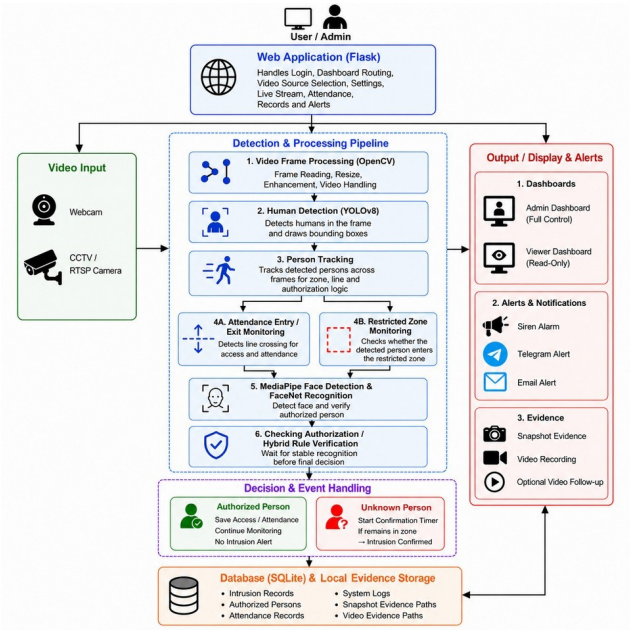



Fig. 2. Intelligent architecture of the proposed CCTV-based AI system.

### *B. Persistent Data and Evidence Design*

SQLite stores structured metadata while snapshots, recorded intrusion videos, authorized-person photos, and face embeddings remain in managed local folders. Intrusion records contain timestamp, event type, duration, status, evidence paths, camera source, and archive metadata. Authorized-person records contain name, role, status, photo and embedding paths, registration date, and removed state. Attendance data is divided into line-crossing events and person-specific sessions containing check-in, check-out, current state, and latest detection. Evaluation samples store the selected condition, ground truth, prediction, timing, and evidence path used to generate metrics.

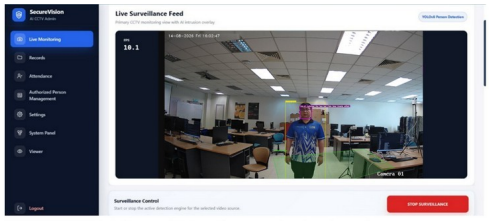
TABLE IV. MAIN PERSISTENT DATA GROUPS

| Data Group | Main Stored Information |
|---|---|
| Intrusion | Time, type, status, snapshot/video, archive |
| Attendance | Identity, direction, check-in/out, state |
| Authorized person | Name, role, photo, embedding, status |
| Settings | Zone, line, recognition, alert parameters |

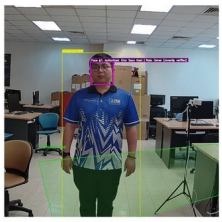
| Data Group | Main Stored Information |
|---|---|
| Evaluation | Condition, truth, prediction, time, evidence |

### *C. Live Monitoring and Human Detection*

The Admin page displays the surveillance feed, video-source selection, restricted-zone and Entry/Exit configuration, detection controls, and system state. YOLOv8 bounding boxes and track information are overlaid on the live frame. The same pipeline supports webcam and CCTV/RTSP input, which permits initial controlled testing without changing the detector. Restricted-zone decisions use the tracked foot point so that the polygon corresponds to a floor area rather than the centre of the body box.



**(a) SecureVision live monitoring interface**

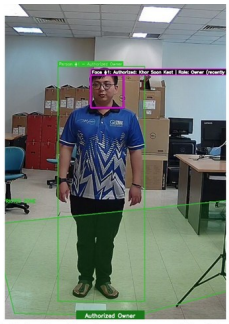


**(b) YOLOv8 human detection and restricted-zone result**

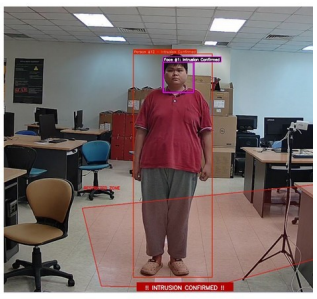
Fig. 3. (a) Live monitoring interface. (b) YOLOv8 detection and restricted-zone result.

### *D. Authorized and Unknown Person Handling*

During face evaluation, the interface displays Checking Authorization and blocks siren, record, and notification actions. When authorization succeeds, the track is labelled with the registered name and role, pending intrusion state is cleared, and monitoring continues without an alert. When the identity remains unknown and the zone confirmation delay is reached, the system activates the siren, captures a snapshot, records video evidence, creates the event record, and sends notifications.



**(a) Authorized person verification**

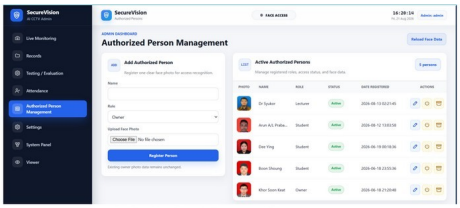


**(b) Unknown person intrusion confirmation**

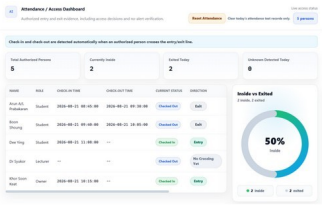
Fig. 4. (a) Authorized verification. (b) Unknown-person intrusion result.

### *E. Management, Records, Alerts, and Readiness*

Authorized Person Management supports profile creation, edit, soft removal, restore, permanent deletion from the removed list, and face-data reload. The Access/Attendance Dashboard summarizes authorized profiles, inside/outside state, and person-specific sessions. Only an authorized track is persisted as an attendance session. History includes date/name/status filtering, pagination, and CSV/Excel export.



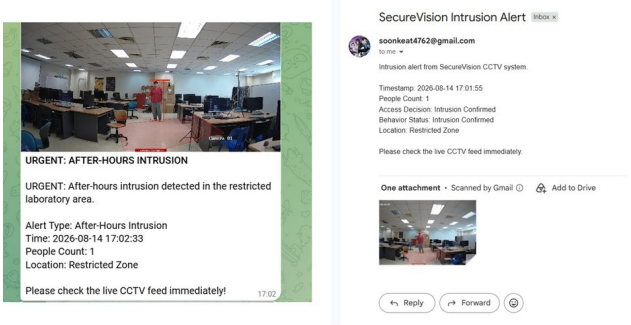
**(a) Authorized Person Management**



**(b) Access / Attendance Dashboard**

Fig. 5. (a) Authorized Person Management. (b) Access/Attendance Dashboard.

Confirmed intrusions are connected to searchable records with evidence preview, workflow status, camera source, archive/restore, pagination, filtered or complete CSV/Excel export, and evidence ZIP packaging. Snapshot alerts are sent immediately through Telegram and email; optional completed-video follow-up is sent only when enabled and within the configured attachment-size limit. Admin control requires authentication, the Viewer is read-only, RTSP credentials are masked, and destructive record actions require confirmation. The System Panel checks CCTV IP, expected Ethernet IP, RTSP port status, and overall network readiness before monitoring starts.




**(a) Telegram intrusion alert** **(b) Email intrusion alert**

Fig. 6. (a) Telegram intrusion alert. (b) Email intrusion alert.

### *F. Implementation Environment and Operational Sequence*

The prototype runs on Python 3.10.11 with TensorFlow 2.15.1, Keras 2.15.0, Ultralytics YOLO, MediaPipe, OpenCV, Flask, and SQLite. The recognition environment uses FaceNet embeddings, while the web process exposes the monitoring and management functions through a local browser interface. Webcam input is used for controlled development and repeatable testing; Hikvision CCTV/RTSP input uses the same video-processing pipeline after network readiness has been verified.

A typical operating sequence is: authenticate as Admin, verify camera/network readiness, select the input source, configure the protected polygon and finite Entry/Exit line, confirm alert settings, and start monitoring. During monitoring, the Viewer role can observe the feed without changing configuration. After an event, the Admin follows the

linked record to inspect evidence and notification status, changes the workflow state from New to Reviewed or Resolved, and archives or exports the material when appropriate. Stop control releases the active monitoring state so that the source can be changed or the application can be closed safely.

## V. TESTING AND EVALUATION FRAMEWORK

### *A. Evaluation Design*

The project contains an original live evaluation dashboard and three controlled modes: Face Angle, Distance, and Movement Speed. The pages reuse the live system prediction instead of introducing a separate evaluation-only AI pipeline. Each saved sample or trial is linked to its condition, ground truth, prediction, processing or detection time, and captured evidence. Result cards, condition summaries, charts, and CSV export are generated from stored labelled data rather than manually entered performance percentages.

TABLE V. CONTROLLED EVALUATION CONDITIONS

| Mode | Conditions | Samples / Evidence |
|---|---|---|
| Face Angle | 0°, ±45°, ±90° | 10 labelled images + time |
| Distance | 1 m to 6 m | 12 labelled images + time |
| Movement | Slow/Normal/Fast | 6 trials, 118 observations |

### *B. Evaluation Metrics*

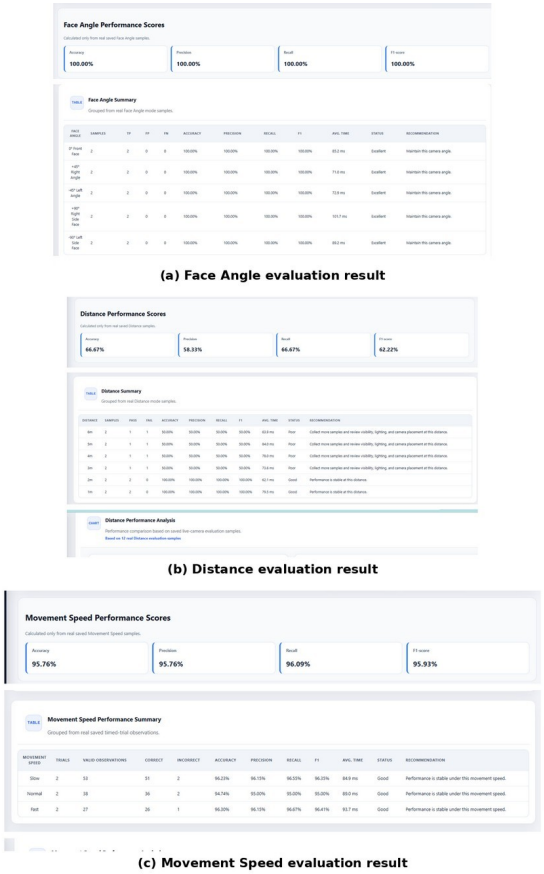
Accuracy measures the overall proportion of correct predictions. Precision is the proportion of predicted positive cases that are correct, Recall is the proportion of actual positive cases detected, and F1-score is the harmonic mean of Precision and Recall. The evaluation page computes:

$$Accuracy = (TP + TN)/(TP + TN + FP + FN) \quad (4)$$

$$Precision = TP/(TP + FP), \; Recall = TP/(TP + FN) \quad (5)$$

$$F1 = 2(Precision \times Recall)/(Precision + Recall) \quad (6)$$

The Face Angle and Distance modes store sample-level ground-truth and prediction labels. Movement Speed aggregates valid detector observations within timed trials. Attendance Entry/Exit is documented separately because its practical reliability also depends on track continuity and physical line calibration.




(a) Face Angle evaluation result

(b) Distance evaluation result

(c) Movement Speed evaluation result

Fig. 7. Evaluation evidence for Face Angle, Distance, and Movement Speed.

### *C. Test Execution and Data Integrity*

For Face Angle, the tester selects one of five configured orientations, captures a frozen evaluation frame, assigns ground truth, and saves the live prediction and detection time. Distance follows the same procedure for 1 m through 6 m. Movement Speed uses timed Slow, Normal, and Fast trials; each trial aggregates repeated valid detector observations and stores duration, correct and incorrect counts, observation total, calculated accuracy, and evidence. Samples may be reviewed, corrected when the ground-truth entry is wrong, deleted, or exported to CSV.

The result pages query the stored samples and recompute aggregate and per-condition metrics. This separation between observation capture and metric calculation prevents the displayed percentages from being typed directly into the interface. Evidence paths retain the image or trial proof associated with each record. The report therefore distinguishes implemented functional verification from quantitative recognition metrics and from physical Attendance line-crossing evidence.

### *D. Functional Verification*

TABLE VI. FUNCTIONAL VERIFICATION SUMMARY

| Function | Integrated Evidence | Status |
|---|---|---|
| Human / zone | YOLOv8 track + foot-point polygon | Done |
| Authorization | Checking state + FaceNet votes | Done |
| Unknown intrusion | Delay + siren + evidence | Done |
| Attendance | Finite line + authorized session | Done |

| Function | Integrated Evidence | Status |
|---|---|---|
| Records | Filter, paging, archive, export | Done |
| Alerts | Telegram/email + video follow-up | Done |

### *E. Reproducibility and Interpretation Rules*

To repeat a controlled test, the operator first confirms that the monitoring pipeline and selected video source are running, chooses the evaluation mode and condition, positions the subject according to that condition, and captures the frozen frame or timed trial. Ground truth is entered before saving the record together with the current prediction, timing, and evidence. Repeating the same condition adds another independent stored sample; it does not overwrite the earlier observation. Aggregate results are regenerated from all retained records after an edit or deletion.

Three interpretation rules are applied. First, a functional status of Done confirms that the integrated workflow exists and produced the expected evidence, but it is not a quantitative accuracy value. Second, high percentages from small controlled samples are reported together with the sample count and must not be presented as broad deployment guarantees. Third, Attendance line-crossing reliability is not merged into Face Angle, Distance, or Movement Speed metrics because it introduces additional variables such as tracking continuity, line placement, and crossing direction.

## VI. RESULTS AND DISCUSSION

### *A. Overall Quantitative Results*

TABLE VII. OVERALL QUANTITATIVE EVALUATION RESULTS (%)

| Mode | Acc. | Prec. | Rec. | F1 |
|---|---|---|---|---|
| Face Angle | 100.00 | 100.00 | 100.00 | 100.00 |
| Distance | 66.67 | 58.33 | 66.67 | 62.22 |
| Movement | 95.76 | 95.76 | 96.09 | 95.93 |

### *B. Face Angle Result*

Ten saved samples covered the five configured angles. All ten were evaluated correctly, producing 100.00% Accuracy, Precision, Recall, and F1-score. Average detection time ranged from 71.0 ms at +45° Right Angle to 101.7 ms at +90° Right Side Face. The result confirms correct operation for this controlled sample set, but the small dataset does not establish broad generalization to different users, cameras, or lighting.

### *C. Distance Result*

Twelve samples covered 1 m to 6 m. The aggregate result was 66.67% Accuracy, 58.33% Precision, 66.67% Recall, and 62.22% F1-score. The 1 m and 2 m conditions recorded 100% accuracy, whereas each condition from 3 m to 6 m recorded 50% accuracy in the current two-sample-per-distance dataset. Distance therefore caused the largest performance reduction, consistent with the smaller and less detailed visible face region at longer range.

### *D. Movement Speed Result*

Six timed trials produced 118 valid observations. Overall performance was 95.76% Accuracy, 95.76% Precision, 96.09% Recall, and 95.93% F1-score. Slow, Normal, and Fast conditions recorded 96.23%, 94.74%, and 96.30% accuracy, respectively. The close condition-level values indicate stable person-detection behaviour across the tested movement speeds in the controlled environment.

### *E. Operational and Record-Management Result*

The integrated workflow handled an intrusion as a structured event rather than a temporary screen message. From a confirmed unknown-person decision, the system could create snapshot and video evidence, save metadata, send alerts, display the record for review, change workflow status, archive and restore it, and export the related data. Authorized access followed a separate path that cleared intrusion state and updated attendance only after identity confirmation. This separation directly supports the project objectives and reduces confusion between access and security events.

### *F. Limitations and Validity*

The measurements represent a laboratory-scale prototype and controlled samples. Face recognition remains sensitive to illumination, pose, blur, occlusion, camera resolution, and distance. Attendance depends on stable track IDs, timely authorization, suitable line placement, and correct direction settings; overlap or track-ID changes can affect an individual crossing. CCTV/RTSP performance additionally depends on network conditions. The Face Angle and Distance sample sizes are small, so the reported percentages demonstrate the implemented evaluation workflow and current configuration rather than production-level accuracy guarantees.

A broader validation should include more participants, repeated sessions, multiple cameras, changing illumination, partial occlusion, and separate train/validation/test identities. Precision of the Entry/Exit subsystem should also be measured through labelled crossing sequences. Future deployment should protect credentials outside the interface, apply stronger account controls, maintain encrypted off-device backups, and formalize retention policies for face data and recorded evidence.

### *G. Achievement of Project Objectives*

Objective 1 is supported by the person-first YOLOv8 and restricted-zone pipeline, which accepts a tracked person and evaluates the foot point against the protected polygon while excluding non-human detections from security decisions. Objective 2 is supported by MediaPipe face localization, FaceNet embedding comparison, repeated identity voting, the Checking Authorization state, and separate authorized/unknown outcomes. Objective 3 is supported by the integrated attendance/access sessions, intrusion evidence, workflow statuses, archive/restore, exports, settings, system readiness, and Telegram/email notification.

The evaluation results also indicate where each objective requires further improvement. Human detection remained comparatively stable under the tested movement speeds, while face recognition degraded as distance increased. Operational integration was functionally verified, but long-term reliability depends on camera placement, calibration, identity diversity, and network conditions. The prototype should therefore be regarded as a completed FYP system and evaluation platform, with further dataset expansion required before broad deployment claims are made.

## VII. CONCLUSION

This paper presented SecureVision, a CCTV-based AI intrusion detection prototype that integrates YOLOv8 person detection and tracking, restricted-zone monitoring, MediaPipe face localization, FaceNet recognition, multi-frame authorization, finite Entry/Exit monitoring, structured SQLite records, snapshot and recorded-video evidence, siren activation, and Telegram/email alerts. Admin and Viewer dashboards support live monitoring, authorized-person management, attendance/access review, records, settings, readiness checking, archive/restore, filtering, pagination, and export.

Controlled evaluation produced 100.00% accuracy for Face Angle, 66.67% accuracy and 62.22% F1-score for Distance, and 95.76% accuracy with 95.93% F1-score for Movement Speed. The results show that recognition distance is the main constraint in the current dataset. Overall, the prototype demonstrates an explainable person-first workflow that distinguishes authorized access from unknown-person intrusion and connects each decision to reviewable evidence and records.

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